

BUSINESS CASE - CLOUD-BASED FACIAL RECOGNITION SYSTEM

Business Name:

SecureVision Technologies Ltd

Industry:

Smart Security & Cloud-Based Access Control Solutions

Executive Summary

SecureVision Technologies Ltd is a UK-based cloud security company providing **next-generation biometric access control systems** to corporate offices, financial institutions, and smart facilities. The company identified a critical gap in traditional access control systems, which rely on physical credentials such as cards and PINs that are prone to theft, sharing, and misuse.

To address this, SecureVision developed a **fully serverless, AI-driven facial recognition system on AWS**, enabling secure, contactless, and real-time identity verification using modern cloud-native architecture.

Problem Statement

Traditional access systems present major limitations:

- Access cards can be lost or stolen
- PIN codes can be shared or compromised
- Manual verification processes are inefficient and error-prone
- Systems lack scalability across multiple locations

This creates:

- Security vulnerabilities
- Operational inefficiencies
- High maintenance costs

Proposed Solution

SecureVision implemented a **cloud-native, serverless biometric system** built on AWS, with the following architecture:

- **Frontend:** React-based web interface for image upload
- **Storage:** Amazon S3 for secure image storage
- **Processing:** AWS Lambda triggered by S3 events
- **AI Engine:** Amazon Rekognition for facial detection and comparison
- **Database:** DynamoDB for storing metadata and results
- **API Layer:** API Gateway for secure communication
- **Security:** IAM roles with least-privilege access

Entire system is:

- Event-driven
- Serverless
- Fully managed
- Automatically scalable

How the System Works (Operational Flow)

1. User uploads image via frontend
2. Image stored in S3 bucket
3. S3 triggers Lambda function
4. Lambda calls Rekognition APIs:
 - DetectFaces
 - CompareFaces
5. Results processed and stored in DynamoDB
6. API Gateway returns response to frontend
7. Access decision made in real time (<1 second)

This ensures **instant, automated identity verification**

Business Value

1. Security Enhancement

- Eliminates impersonation risks
- Biometric identity is unique
- No dependency on passwords or cards

2. Operational Efficiency

- No manual checks required
- Real-time access decisions
- Reduced security staff workload

3. Cost Optimisation

- Serverless → pay-per-use model
- No infrastructure maintenance
- Eliminates hardware dependency

4. Scalability

- Seamlessly scales from:
 - 10 users → 10,000+ users
- Handles peak traffic automatically

5. User Experience

- Fully contactless system
- Fast and seamless access
- Modern digital workplace solution

DevOps & Cloud Engineering Value

This solution demonstrates:

- Event-driven architecture
- Serverless microservices design
- AI/ML integration into cloud workflows
- Secure IAM-based access control
- Real-time API-driven systems
- CloudWatch monitoring and resilience

This is a **production-grade cloud system**

Risks and Mitigation

Risk	Mitigation
False positives	Adjust Rekognition similarity thresholds
Poor image quality	Enforce capture standards
Privacy concerns	Encryption + IAM + secure policies
Latency	Deploy in regional AWS zones
Data security	Strict IAM and bucket policies

Return on Investment (ROI)

- Reduced manpower costs
- Eliminated hardware maintenance
- Faster onboarding/offboarding
- Improved compliance and auditability
- Scalable across multiple office locations

Break-even achieved quickly for medium to large enterprises

Real-World Impact

Within months of deployment:

- Operational costs reduced significantly
- Security incidents minimised
- Client adoption increased due to:
 - reliability
 - scalability
 - automation

Industry Relevance

This architecture is widely applicable across:

- Corporate security systems
- Banking (KYC verification)
- Retail (visual search)
- Healthcare (patient identification)
- Logistics (identity validation)

It represents a **modern standard for cloud-based AI systems**

Final Assessment

SecureVision Technologies Ltd successfully implemented:

- Serverless cloud architecture
- AI-powered identity verification
- Fully automated DevOps workflows
- Scalable and secure infrastructure

This is a **real-world, enterprise-grade cloud engineering solution** designed for modern digital environments.